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On the Origins of the Block Structure Phenomenon in Neural Network Representations

Thao Nguyen* Maithra Raghu† Simon Kornblith†

Abstract

Recent work [29] has uncovered a striking phenomenon in large-capacity neural networks: they contain blocks of contiguous hidden layers with highly similar representations. This *block structure* has two seemingly contradictory properties: on the one hand, its constituent layers exhibit highly similar dominant first principal components (PCs), but on the other hand, their representations, and their common first PC, are highly dissimilar across different random seeds. Our work seeks to reconcile these discrepant properties by investigating the origin of the block structure in relation to the data and training methods. By analyzing properties of the dominant PCs, we find that the block structure arises from *dominant datapoints* — a small group of examples that share similar image statistics (e.g. background color). However, the set of dominant datapoints, and the precise shared image statistic, can vary across random seeds. Thus, the block structure reflects meaningful dataset statistics, but is simultaneously unique to each model. Through studying hidden layer activations and creating synthetic datapoints, we demonstrate that these simple image statistics dominate the representational geometry of the layers inside the block structure. We explore how the phenomenon evolves through training, finding that the block structure takes shape early in training, but the underlying representations and the corresponding dominant datapoints continue to change substantially. Finally, we study the interplay between the block structure and different training mechanisms, introducing a targeted intervention to eliminate the block structure, as well as examining the effects of pretraining and Shake-Shake regularization.

1 Introduction

Many modern successes of deep neural networks have adopted simple techniques to systematically increase model capacity, often through scaling architecture depth and width [36]. These large capacity models typically also maintain strong performance even in tasks with small amounts of training data. This has led to their widespread use across different many applications, including data-scarce, high-stakes settings such as medical imaging [22, 40].

However, recent work has shown that the representational structures of these large capacity models exhibit distinctive properties that are not present in shallower/narrower networks. Specifically, when using linear centered kernel alignment (CKA) [19] to measure similarity between hidden representations of neural network layers, Nguyen et al. [29] show that a large set of contiguous layers that share highly similar representations. This is visible as a clear *block structure* in the heatmap of

*University of Washington, thaottn@cs.washington.edu. Work done as part of the Google AI Residency.
†Google Brain, {maithra,skornblith}@google.com

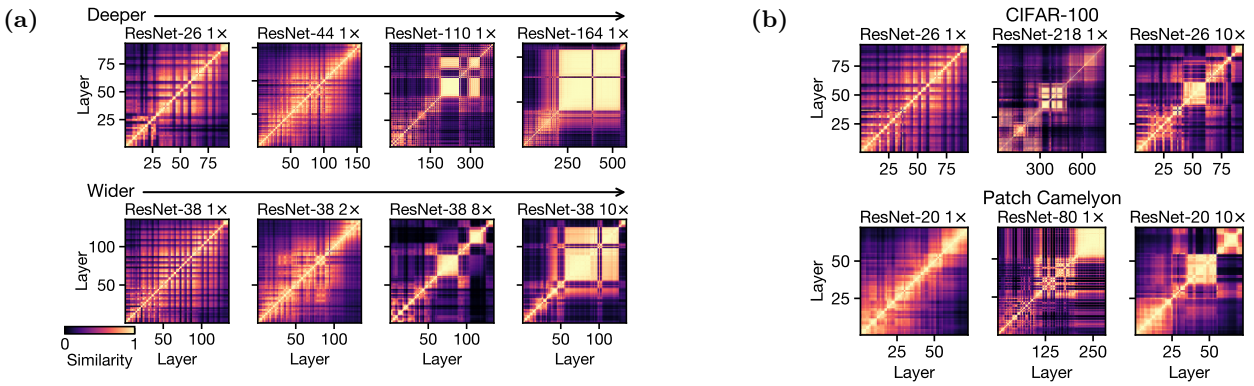

Figure 1: (a): Block structure arises in wide and deep networks. Heatmaps show similarity between layer representations, measured with linear CKA, for models of varying widths and depths trained on CIFAR-10. As models become wider or deeper, "blocks" of consecutive layers have highly similar representations. **(b): Block structure arises on many datasets.** Models trained on CIFAR-100 and the medical histopathology dataset Patch Camelyon show similar behavior to those in panel (a). Block structure also appears when networks trained on CIFAR-10 are evaluated on OOD images; see Appendix B.

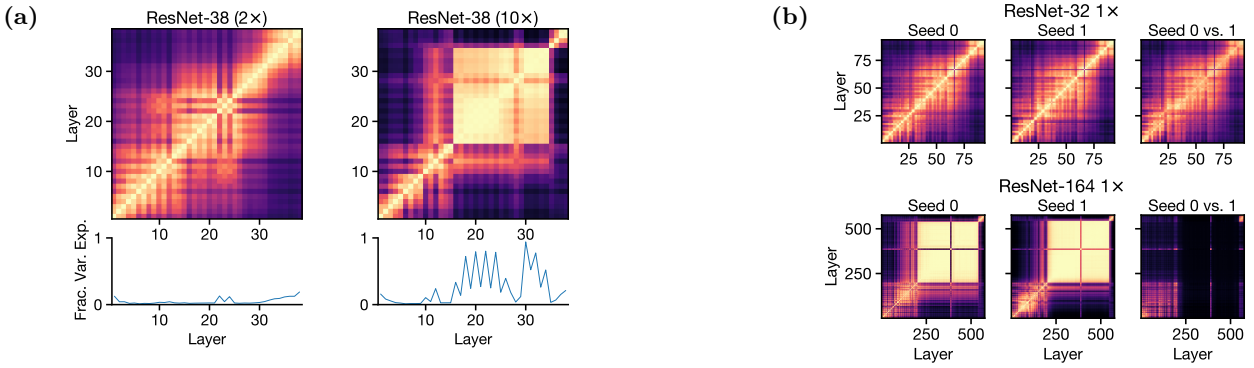

Figure 2: (a): The first principal component of representations inside the block structure explains the majority of the variance in representations. Top: Linear CKA similarity heatmaps for networks that do and do not exhibit block structure. Bottom: Fraction of variance in representations explained by first principal component. See Appendix C for more networks. **(b): Representations inside the block structure are highly unique to each seed.** 2 leftmost panels show similarity of representations *within* networks trained with 2 different random seeds. Rightmost panel shows similarity *between* the 2 networks.

pairwise linear CKA similarities between layers (see Figure 1a). This block structure phenomenon is robust, appearing both in models trained on natural image datasets and those trained on medical imaging datasets (Figure 1b), as well as in a variety of CNN architectures, such as those with and without residual connections [29].

Despite being so consistently prevalent, the block structure exhibits strangely contradictory properties. On the one hand, it arises from measurable properties of learned data representations. Specifically, Nguyen et al. [29] find that a block structure is present if and only if its constituent layer representations have a dominant first principal component (PC), which is in fact shared among them (Figure 2a). This suggests that the block structure (and the common first PC) capture key attributes of the underlying dataset. But on the other hand, this first PC is highly *dissimilar* across random seeds (Figure 2b). And such representational inconsistencies have been linked to overfitting

to spurious data features and poor generalization [26].

These findings thus motivate a pressing fundamental question — is the block structure a sign of overfitting to idiosyncrasies of the data and training process, or does it pick up meaningful signals? In this paper, we uncover answers to this question, exploring the origin of the block structure in relation to the data and training methods, and reconciling its contradictory behaviors. Specifically, our results are as follows:

- The dominant first principal components of the layers that make up the block structure arise from a small number of *dominant datapoints* that share similar characteristics (e.g., background color). Like the block structure phenomenon, dominant datapoints are found in large-capacity neural networks.
- The set of dominant datapoints (and their shared characteristics) can vary across training runs, which explains the observed block structure dissimilarities across seeds.
- The block structure is caused by these dominant datapoints. When dominant datapoints are excluded from the dataset used for representational analysis, the block structure disappears.
- Dominant datapoints produce high activation norms inside the hidden layers corresponding to the block structure. By constructing synthetic examples based off of the shared image characteristics of the dominant datapoints, we show that these image characteristics are indeed responsible for the high activation norms.
- The block structure emerges early in training, but early block structures have different representations and yield different dominant datapoints than the final block structure at the end of training.
- We show that it is possible to eliminate the block structure without interfering with generalization using a novel principal component regularization method. We also show that alternative training mechanisms such as transfer learning and Shake-Shake regularization can reduce the block structure and yield more consistent representations across different training runs.

2 Related Work

Previous work has studied certain propensities of deep neural networks in the standard training setting, such as their simplicity bias [17, 27, 38], texture bias [2, 8, 14] and reliance on spurious correlations [9, 15, 18, 24, 33]. Inspired by the findings of Nguyen et al. [29] that deep and wide networks learn many layers with similar representations, we seek to characterize what signals these layers convey and may be overfitting to. To do so, we analyze the behavior of the *internal representations* of models of varied depths and widths, using methods for measuring similarity of neural network hidden representations [19, 26, 30], as well as standard tools of linear algebra. Representational similarity techniques have previously shed light on model training procedures [10, 28], features [13, 32, 37], and dynamics [23], and has also furthered understanding of network internals in applications of deep learning, such as medicine [31] and machine translation [4].

Our work also relates to previous attempts to understand the properties of overparameterized models [42]. Theoretical work in this area has focused on linear models, models with random features, or kernel settings [3, 5, 11, 21], all of which lack intermediate features, or involves linear networks [1].